Reuben Abraham T. Georgi

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Robotics engineer with 3 years of experience and a Master's degree in Control and Autonomy. Experience and interests centre on classical and learning-based techniques for control, planning, and perception. Striving to develop safe and reliable autonomous systems for real-world deployment.

EDUCATION

Purdue University West Lafayette, USA
Master of Science in Aeronautics and Astronautics (MSAA) May 2021

Punjab Engineering College Chandigarh, India
Bachelor of Technology (B.Tech) in Aerospace Engineering October 2019

WORK EXPERIENCE/INTERNSHIPS

Simelabs (Astek Group) Kochi/Bangalore, India
Robotics System Engineer February 2022 - Present

Autonomous Mobile Robot Demo with Vision-Based Decision Making

- Developed a TurtleBot3-based demo using camera-based perception (OpenCV & trained Keras models) and Nav2 to interpret environmental cues (signals, signage, timed crossing gates) and make navigation decisions
- Investigated camera latency and odometry-related localization issues, and delivered a system usable via a custom PyQt testing GUI or an external Android application

R&D Automation Project (At **West Pharmaceutical Services**)

- Developed proof of concept projects for:
  - ***Pose estimation for autonomous pick-and-place*** using an Intel RealSense depth camera and ABB YuMi, leveraging Open3D (RANSAC & ICP), YOLO object detection, SAM segmentation, and socket communication, achieving ~10 mm pose accuracy at long working distances
  - ***Ensuring collision-free trajectories during assembly line restarts*** using MoveIt 2 and ROS 2 to reduce manual intervention and programming efforts
  - ***Remote visualization of assembly operations in real-time*** using RobotStudio with robot and PLC data
- Simulated and programmed ABB (YuMi, GoFa) and Omron (TM5) robots for process checking, layout validation, cycle time optimization and assembly operations in RoboDK and RobotStudio

AbbVie Inc. Chicago, USA
Research Analyst/ Summer Intern July - September 2020

- Conducted CFD simulations in M-Star CFD and Ansys to characterize the hydrodynamics of USP dissolution apparatus with results published in the Journal of Pharmaceutical Sciences

Helicopter Division, Hindustan Aeronautics Limited Bangalore, India
Intern January - May 2018

- Internship Project: Mathematical modelling of helicopter blades to simulate response and predict performance of an isolated rotor

SPACE lab, Khalifa University Abu Dhabi, U.A.E
Intern May - July 2017

- Assembled and programmed TETRIX Ranger using MATLAB and Arduino for line following tasks and set up a Pumpkin CubeSat kit using MPLAB

SELECTED PROJECTS

UAV Simulation and Motion Planning Framework

- Developing a modular and extensible Python tool for simulating quadcopter motion and experimenting with different motion planning and control techniques
- Currently implemented features:
  - Simulation: Set actuator limits, actuator delays, waypoints, wind disturbances (constant or Gaussian), obstacles (spherical)
  - Planning: Dijkstra, A*, RRT*, iLQR (Iterative Linear Quadratic Regulator)
  - Control: Cascade PID, LQR (using JAX for autodiff), linear MPC (using CasADi for optimization)
  - Visualization: Plots of state variable evolution with time, animation of quadcopter flight

Autonomous Mobile Manipulator Simulation

- Designed and implemented a ROS 2 mobile manipulator system integrating Nav2, MoveIt 2, and Gazebo to enable autonomous navigation and manipulation
- Utilized BehaviorTree.CPP to react to dynamically spawned objects and orchestrate task-level actions

3D SLAM

- Implemented the extended and unscented Kalman filter (EKF, UKF) in Python for simultaneous localization and mapping in 3 dimensions
- Simulated scenarios with known correspondences and unknown number of landmarks and introduced uncertainty by adding noise to odometry and sensor measurements

Quadcopter Design using Genetic Algorithms

- Formulated a multi-objective problem to minimize cost and weight of a quadcopter subject to performance and integration constraints
- Developed MATLAB app for users to provide data and to visualize results, including the Pareto frontier which displays Pareto optimal solutions.

TECHNICAL SKILLS

Languages: **MATLAB**, **Python**, **C++**

Packages/Libraries: **Nav2 (ROS 2), MoveIt 2 (ROS 2), Gazebo, NumPy, Matplotlib, OpenCV, JAX, BehaviorTree.CPP**

Other software: **ABB RobotStudio/RAPID**, **RoboDK**, **Simulink, ANSYS**

PUBLICATIONS

Alexander M. Kubinski, Gayathri Shivkumar, Reuben A. Georgi, Susan George, James Reynolds, Ricardo D. Sosa, Tzuchi R. Ju, **Predictive Drug Release Modeling Across Dissolution Apparatuses I and II using Computational Fluid Dynamics,** Journal of Pharmaceutical Sciences, Volume 112, Issue 3, 2023, Pages 808-819, ISSN 0022-3549, https://doi.org/10.1016/j.xphs.2022.10.027.

Alexander M. Kubinski, Ricardo D. Sosa, Gayathri Shivkumar, Reuben Georgi, Susan George, Eric J. Murphy, Tzuchi R. Ju, **Predictive dissolution modeling across USP apparatuses I, II, and III**, Journal of Pharmaceutical Sciences, Volume 114, Issue 6, 2025, 103765, ISSN 0022-3549, https://doi.org/10.1016/j.xphs.2025.103765.